THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET B

Chapters 5–13: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity, Our Environment** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. In humans, the number of chambers in the heart is: 1

(a) 1 **(b)** 4 **(c)** 3 **(d)** 2

2. The energy currency of the cell is: 1

(a) NADP **(b)** ADP **(c)** ATP **(d)** glucose

3. Tendrils of a pea plant coil around a support due to: 1

(a) thigmotropism **(b)** chemotropism **(c)** phototropism **(d)** hydrotropism

4. The spinal cord is protected by the: 1

(a) pelvis **(b)** skull **(c)** vertebral column **(d)** ribs

5. Testosterone is produced by: 1

(a) adrenal **(b)** ovaries **(c)** thyroid **(d)** testes

6. The tube that carries both urine and sperm in males is the: 1

(a) ureter **(b)** oviduct **(c)** vas deferens **(d)** urethra

7. The site of fertilisation in human females is the: 1

(a) vagina **(b)** ovary
(c) fallopian tube (oviduct) **(d)** uterus

8. The male reproductive part of a flower is the: 1

(a) sepal **(b)** stamen **(c)** pistil **(d)** petal

9. In the F2 of a dihybrid cross, the number of plants with both recessive traits out of 16 is: 1

(a) 9 **(b)** 1 **(c)** 0 **(d)** 3

10. A cross between a tall (TT) and a short (tt) pea plant gives F1 plants that are: 1

(a) all tall **(b)** medium height
(c) all short **(d)** half tall, half short

11. For the chemicals that deplete the ozone layer, Student A says CFCs; Student B says carbon dioxide. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

12. In a food chain, the organisms that make food from sunlight are: 1

(a) decomposers **(b)** producers **(c)** carnivores **(d)** herbivores

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Why do aquatic animals breathe faster than terrestrial animals? 2

14. Name two plant hormones that promote growth. 2

15. A student says that the uterus prepares itself every month to receive a fertilised egg. Justify or refute. 2

16. What is meant by a monohybrid cross? 2

17. What is an ecosystem? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. What is dialysis? Why is it needed? 3

19. How does chemical coordination take place in animals? Give two examples. 3

20. What is the importance of variation in a population? (three) 3

21. How do our habits affect the environment with respect to waste? Give two changes we can make. 3

22. What will happen if we kill all the organisms in one trophic level? 3

OR

Suggest three ways to reduce the problem of waste disposal. 3

23. How do plants get rid of their excretory products? 3

OR

Explain double circulation in humans. Why is it necessary? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Explain the mode of nutrition in plants. 5
(ii) Describe the role of stomata in photosynthesis and transpiration.
(iii) How do desert plants manage?

OR

(i) Describe the human respiratory system with a labelled diagram (main parts). 5
(ii) How are the lungs designed to maximise gas exchange?

25. (i) Describe the human male reproductive system. 5
(ii) Why are testes located outside the abdominal cavity?
(iii) Function of vas deferens.

OR

(i) Explain three methods of asexual reproduction with examples. 5
(ii) Why is asexual reproduction considered faster?

END OF QUESTION PAPER — Total: 50 Marks